

Module-02

Software Development Life Cycle (SDLC) in the Age of AI

Q1. Draw the SDLC lifecycle diagram and briefly explain the workflow.

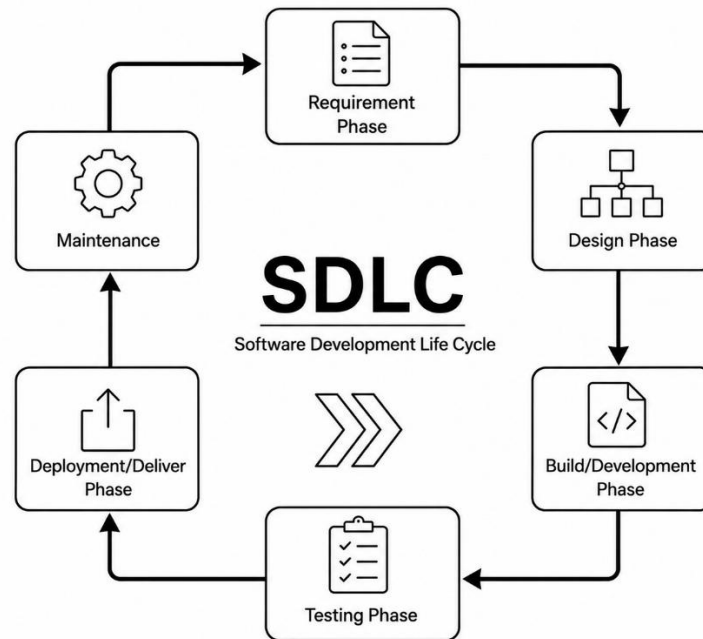


Fig1: Software Development Life Cycle (SDLC).

1. Requirement Analysis Phase: Requirement Analysis is the initial and one of the most crucial phases of SDLC. In this stage, the development team will talk with the stakeholders to define a customer's desired features and functions. They create Business Requirements Document (BRD), Software Requirements Specification (SRS), and review and approve Software and Business requirements. They meet with stakeholders and clients, gather business requirements, and define functional and non-functional requirements and system constraints. These documents are "BRD", "SRS", "Use Cases" and "Requirement Traceability Matrix (RTM)".

2. Design Phase: Once the requirements have been approved, the solution will be designed. Systems are built during the Design Phase, which specifies the design of the system. Developers are not just coding right away – first they develop architecture and technical blueprint. Some Activities are High Level Design (HLD), Low Level Design (LLD), Database Design, API Design, UI/UX Design. HLD is a high-level overview of the overall system, and is prepared by the Solution or System Architect. It is utilized by the Architect, Business Analyst, Project Manager and QA. It contains architecture and modules and there are questions on how the system is designed. LLD is all about specific modules with detailed designs and it is prepared by the Technical Lead or Senior Developer. It is used by Developer / QA. It contains any APIs, logic, database or answers to how the module is implemented.

3. Implementation Development Phase: The Implementation Phase involves creating real software from the approved design, making use of coding and programming of source code. Coding, code review, unit testing, version controlling with git, integration testing, CI/CD integration, etc. are activities that are covered.

4. Testing Phase: Testing verifies that the software meets all the specifications agreed upon and is devoid of any crucial bugs before its launch. Activities involved in it are Test planning, Test case design, Test execution, Defect Reporting, Regression testing, Retesting, Automation testing, Performance testing, Security testing, User Acceptance Testing (UAT).

5. Deployment Phase: App is tested, approved and deployed to the production environment for the end-users. Production release, database migration, configuration changes, smoke testing, monitoring and roll back planning.

6. Maintenance Phase: Once deployed, maintenance starts and is followed throughout the software life cycle to assure that the software application remains functional, secure and current. Examples of activities are bug fixes, performance enhancement, feature enhancements, security patch fixes, customer support and monitoring. There are four types of maintenance. Corrective Maintenance is for bugs which are discovered after release. Adaptive user's experience maintenance brings new OS and hardware support capabilities. Perfective Maintenance introduces new features or enhances old features, based on user feedback. Preventive Maintenance enhances code quality, infrastructure, to minimize future issues.

Q2. Practical AI Exercise

Table-1: Five user stories for an E-commerce application (Reviewed and Improved Manually).

Story ID	User Story	Priority
US-01	As a shopper, I want to browse products by category, so that I can quickly find products relevant to my needs.	High
US-02	As a shopper, I want to search for products using keywords, so that I can quickly find specific products.	High
US-03	As a shopper, I want to add products to my shopping cart, so that I can purchase multiple products in a single checkout.	High
US-04	As a shopper, I want to select a payment method and complete my payment securely, so that I can successfully place my order.	High
US-05	As a shopper, I want to view the current status of my order, so that I can track its delivery progress.	Medium

Acceptance Criteria

(Reviewed and Improved Manually)

Selected User Story: US-04

As a shopper, I want to select a payment method and complete my payment securely, so that I can successfully place my order.

AC 1 - Payment Method Selection

Given I am on the checkout page with items in my cart,

When I select an available payment method,

Then the selected payment method should be clearly indicated and saved for the current order.

AC 2 - Successful Payment

Given I have entered valid payment information,

When I click “Pay Now”,

Then the system should process the payment and display the payment result within 10 seconds.

AC 3 - Invalid Payment Information

Given I have entered invalid or incomplete payment information,

When I click “Pay Now”,

Then the system should display a clear validation error and prevent the order from being placed.

AC 4 - Order Confirmation

Given the payment is successful,

When the transaction is completed,

Then the system should create the order and display an order confirmation containing a unique order ID.

AC 5 - Payment Failure

Given the payment fails because of a network or payment gateway error,

When the failure occurs,

Then the system should display an appropriate error message and keep the cart items available for another payment attempt.

AC 6 - Secure Payment Communication

Given I am submitting an online payment,

When payment information is transmitted between the application and server,

Then the communication should use HTTPS to protect the transmitted data.

Table-2: Sprint Backlog (Reviewed and Improved Manually).

Task ID	User Story	Task Description	Priority	Story Points	Status
T-01	US-01	Design and implement product category listing page	High	3	To Do
T-02	US-01	Implement category filtering and product display logic	High	2	To Do
T-03	US-02	Implement keyword-based product search	High	3	To Do
T-04	US-02	Implement search results page with pagination	High	5	To Do
T-05	US-03	Implement Add to Cart functionality	High	3	To Do
T-06	US-03	Implement shopping cart with update and remove options	High	5	To Do
T-07	US-04	Integrate the selected payment gateway API	High	8	To Do
T-08	US-04	Implement checkout page with payment method selection	High	5	To Do
T-09	US-04	Implement payment success, failure, and retry handling	High	3	To Do
T-10	US-04	Verify secure HTTPS communication for payment requests	High	3	To Do
T-11	US-05	Design and implement order tracking page	Medium	3	To Do
T-12	US-05	Connect order status updates with the backend/database	Medium	5	To Do

What the AI did well.

1. Produced relevant user stories for E-commerce in the appropriate format.
2. Spearheaded important areas like product viewing, searching, shopping carts, payment and order tracking.
3. Produced more than the necessary 5 acceptance criteria.
4. Utilized the Given-When-Then format in the acceptance criteria.
5. Conducted both positive payments and negative payments.
6. Developed a Sprint Backlog that has task ID, Priority, Story Points and Status.
7. Ensured a good start to subsequent SQA review.

What corrections did you make?

1. Eliminated using the term 'a reasonable time' and substituted a measurable requirement in its place.
2. Refined the user stories to be more specific and clearer.
3. Enhanced the 'Payment' user story to concentrate on payment security and successful order placements.
4. Clarified acceptance criteria and made them measurable.
5. Clarified what to do when a payment fails, and what messages to expect in the event of a failure.
6. Made more specific the security requirement by adding HTTPS.
7. Created an independent Sprint Backlog task for checking secure payment communication.
8. Enhanced task descriptions making them more actionable.
9. Insured the Sprint Backlog is correctly matched against the user stories and acceptance criteria.

Why human review is still necessary.

The output from Artificial Intelligence is often vague, incomplete, non-testable, so human review needs to occur to supplement. AI might have incomplete knowledge of the project's current business practices, technical requirements, security measures, and user needs. A SQA engineer with a human touch can help detect these situations and help make sure that the requirements are clearly defined, complete, measurable, testable, realistic and meet the real project needs.

Screenshots of AI Interaction

The screenshot shows the Claude AI chat interface. The browser tabs include 'Ostad - Learn Skills Live' and 'E-commerce SQA exercise: user stories and sprint backlog'. The URL is 'claude.ai/chat/4a271551-15e4-4ff9-ba27-123773472c8d'. The chat history shows a previous message from the user asking for five user stories in a specific format. The AI's response is displayed in a light blue box, titled 'Task 1: User Stories'. Below the title is a table with three columns: 'Story ID', 'User Story', and 'Priority'. The table contains five rows of user stories. At the bottom of the chat area, there is a text input field with the placeholder 'Write a message...' and a 'Send' button. The system status bar at the bottom shows '32°C Partly sunny' and the time '4:33 PM 8/14/2026'.

Suppose you are a professional SQA engineer. I am completing a practical AI exercise for an E-commerce application. Perform ONLY the following three tasks. Do not review, critique, or improve your own output. I will manually review and improve the generated output afterward.

Task 1: Five User Stories

Create exactly five user stories using the standard format: "As a [user], I want [goal], so that [benefit]."

Make the stories realistic and relevant to common E-commerce features such as product browsing, product search, shopping cart, checkout, payment, and order tracking.

Show more

Task 1: User Stories

Story ID	User Story	Priority
US-01	As a shopper, I want to browse products by category, so that I can quickly find items I'm interested in.	High
US-02	As a shopper, I want to search for products by keyword, so that I can find specific items without browsing manually.	High
US-03	As a shopper, I want to add products to a shopping cart, so that I can purchase multiple items in a single checkout.	High
US-04	As a shopper, I want to securely pay for my order using a payment method of my choice, so that I can complete my purchase confidently.	High
US-05	As a shopper, I want to track the status of my order, so that I know when it will be delivered.	Medium

Write a message...

Sonnet 5 Medium

Claude is AI and can make mistakes. Please double-check responses.

The screenshot shows the Claude AI chat interface. The browser tabs include 'Ostad - Learn Skills Live' and 'E-commerce SQA exercise: user stories and sprint backlog'. The URL is 'claude.ai/chat/4a271551-15e4-4ff9-ba27-123773472c8d'. The chat history shows a previous message from the user asking for acceptance criteria and a sprint backlog. The AI's response is displayed in a light blue box, titled 'Task 2: Acceptance Criteria'. Below the title, it lists the selected user story and then provides six acceptance criteria. At the bottom of the chat area, there is a text input field with the placeholder 'Write a message...' and a 'Send' button. The system status bar at the bottom shows '32°C Partly sunny' and the time '4:33 PM 8/14/2026'.

Task 2: Acceptance Criteria

Selected User Story: US-04 — As a shopper, I want to securely pay for my order using a payment method of my choice, so that I can complete my purchase confidently.

Acceptance Criterion 1:
Given I am on the checkout page with items in my cart, When I select a payment method (e.g., credit card, mobile banking, cash on delivery), Then the selected method should be highlighted and saved for the current order.

Acceptance Criterion 2:
Given I have entered valid payment details, When I click "Pay Now," Then the system should process the payment and display a confirmation message within a reasonable time.

Acceptance Criterion 3:
Given I have entered invalid or incomplete payment details, When I click "Pay Now," Then the system should display a clear error message and prevent the order from being placed.

Acceptance Criterion 4:
Given my payment is successful, When the transaction completes, Then an order confirmation with an order ID and receipt should be generated and shown to me.

Acceptance Criterion 5:
Given my payment fails due to a network or gateway error, When the failure occurs, Then the system should notify me and keep my cart items intact so I can retry payment.

Acceptance Criterion 6:
Given I am on the payment page, When the page loads, Then all payment data fields (e.g., card number) must be transmitted securely (e.g., HTTPS/encryption).

Task 3: Sprint Backlog

Write a message...

Sonnet 5 Medium

Claude is AI and can make mistakes. Please double-check responses.

Ostad - Learn Skills Live

E-commerce SQA exercise: user

Ask Gemini

claude.ai/chat/4a271551-15e4-4ff9-ba27-123773472c8d

E-commerce SQA exercise: user stories and sprint backlog

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Given I am on the payment page, When the page loads, Then all payment data fields (e.g., card number) must be transmitted securely (e.g., HTTPS/encryption).

Task 3: Sprint Backlog

Task ID	User Story	Task Description	Priority	Story Points	Status
T-01	US-01	Design and implement product category listing page	High	3	To Do
T-02	US-01	Create category filter component (UI + logic)	High	2	To Do
T-03	US-02	Implement search bar with keyword-based query	High	3	To Do
T-04	US-02	Add search results page with pagination	High	5	To Do
T-05	US-03	Build "Add to Cart" functionality	High	3	To Do
T-06	US-03	Create shopping cart page (view/update/remove items)	High	5	To Do
T-07	US-04	Integrate payment gateway API	High	8	To Do
T-08	US-04	Build checkout page with payment method selection	High	5	To Do
T-09	US-04	Implement payment success/failure handling	High	3	To Do
T-10	US-05	Design order tracking page (status display)	Medium	3	To Do
T-11	US-05	Connect order status updates backend/database	Medium	5	To Do

Write a message...

Sonnet 5 Medium

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32°C
Partly sunny

4:34 PM
8/14/2026